

MATH 572 Final Project

Modelling Seasonal Influenza Data in Alberta

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1 Abstract

Epidemic modelling is the field of modelling the spread of disease via tools from multiple disciplines such as mathematics, statistics, and computer science (1). It uses a combination of data and prior hypotheses, which can be used to counter-factually analyse the past, model the present, or forecast future trajectories of an epidemic given various scenarios (2). The purpose of this report is to outline the results of the age-structured SIR modelling project on the influenza data from Alberta Health Services.¹ We used a Bayesian method approach to model the 2016-2017 seasonal influenza data on the Albertan population, and conducted a counterfactual analysis assuming two different vaccine effectiveness with various amounts of coverage within each population. Our focus was to assess what could have happened if there were preventative measures in place such as having efforts in place for a higher vaccination coverage. It was found that by maximizing vaccine coverage and efficiency, the number of cumulative cases within group 1 and group 2 decreased by 45.11% and 6.89%, respectively. This project will aid individuals in understanding population dynamics during periods of an influenza infection.

2 Lay Audience

The intended audience of this report is the Government of Alberta - Ministry of Health.

3 Introduction

Influenza viruses have been extensively studied since the discovery of the *Orthomyxoviridae* family of RNA viruses in 1933 (3). Influenza, more commonly known as “the flu”, is a highly contagious respiratory illness which typically spreads from individual to individual via droplets produced when an infected person coughs, sneezes, or talks (4). Although influenza has been extensively studied over the last century, due to the COVID-19 pandemic, heightened awareness of infectious diseases and the importance of preventing their spread has allowed individuals to research the risks and consequences of the influenza virus.

¹ All code used to create plots and complete analyses can be provided upon request at sp8@ualberta.ca.

The so-called susceptible, infectious, and recovered (SIR) model, first proposed by Kermack and McKendrick in 1927, is a deterministic epidemiological model that describes the dynamics of infectious diseases within a population (5). This model describes the behaviour between individuals within a given population. This is a basic model, and since its discovery, many advancements of the SIR model have been made. One such advancement is the age-structured SIR (A-S SIR) model. The A-S SIR model is used to model the spread of infectious diseases within a population with different age groups which allows researchers to consider age-specific parameters of the mathematical model. For this report, we will be advancing the aforementioned A-S SIR model further into adding an additional confirmed compartment, thus building an A-S SICR model. Furthermore, a counterfactual analysis will be conducted on the influenza data to assess what the situation could have been if there were intervention methods that were placed during the season.

Applying these mathematical models to real-world data collected by sources including government health organizations may allow us to learn not only the population dynamics during periods of infection, but also to develop strategies for influenza prevention. The Government of Alberta's health branch collects data on confirmed influenza cases each season from various sources, and has the data available for individual access upon request. We can model the Albertan population to learn and revise different control measures on influenza.

In this research, seasonal influenza data from Alberta Health Services is used to fit the aforementioned A-S SICR model. The objective of this study is to determine a baseline estimate using Bayesian inference for parameter estimation (6), and conduct a counterfactual study by adding an additional vaccination component to the age group < 15 to see whether or not vaccination rates will reduce/increase the number of infectious individuals.

4 Methods

4.1 Description of Data

Data has been made available courtesy of Dr. Betsy Varughese and Alberta Health Services.² Data from a 52 week influenza season in 2016-2017 includes the following factors: season, week start date, week end date, flu week, order, age group, and count. Some counts are indicated as < 5 , therefore a random number generator was used to fill a value to these cells. Unfortunately, a data dictionary was unavailable at hand. Furthermore, it is important to note that influenza is often underreported, and the data collected will cover a fraction of the actual influenza cases in Alberta. Age will be divided into two groups, 0 – 14 (Group 1) and 15+ (Group 2).

The following plot shows weekly cases from the seasonal influenza data:

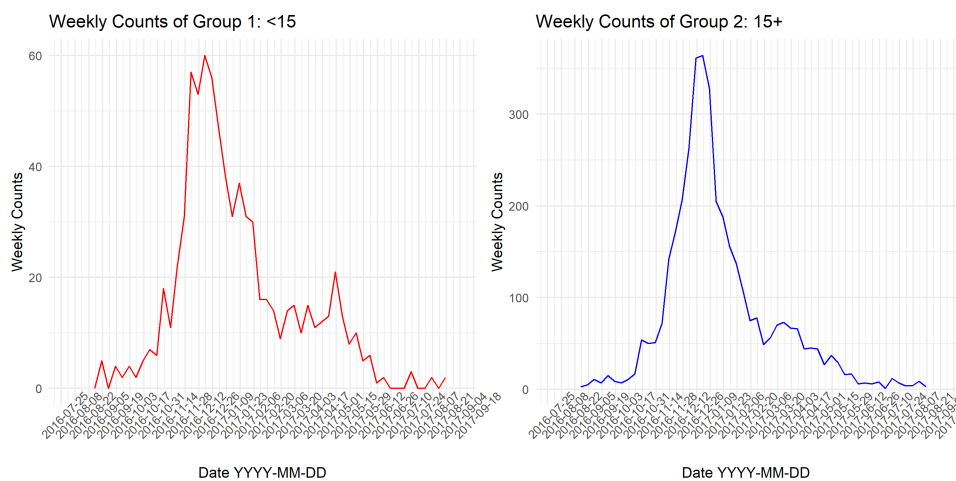


Figure 1: Weekly Counts of 2016-2017 Influenza Data, Left: Group 1, Right: Group 2

As seen in figure 1, cases for both groups peaked in Nov/Dec 2016. For group 1, we can see that there is a peak in cases around April 2017. Weekly counts of group 2 may be seen as significantly higher compared to group 1, but is important to note that this is due to the disproportionality in demographics between age groups in our dataset as seen in figure 10.

²Please contact the author (sp8@ualberta.ca) AND Dr. Betsy Varughese (mvarughe@ualberta.ca) for further information on this data.

4.2 Model Diagram

4.3 Age-structured SICR Model

The age-structured SICR model can be summarized by the compartmental diagram shown in figure 2. The corresponding system of ordinary differential equations (ODEs) can be found in the appendix.³

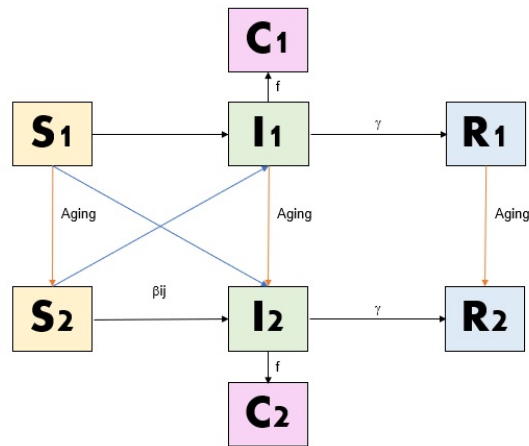


Figure 2: Diagram of Age Structured SICR model compartments

For the counterfactual analysis, the compartmental diagram was adjusted, as seen in the following figure:

³This model was constructed from Dr. Betsy Varughese's Math 572 lecture material.

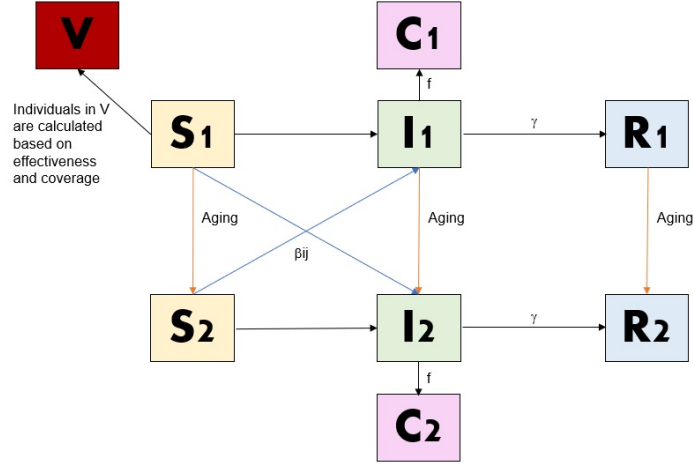


Figure 3: Diagram of Age Structured VSICR model compartments

Cumulative counts of confirmed case compartments from the SIRC model were compared to the respective compartments of the VSICR model to determine the percent increase/decrease of varying vaccine efficiency and coverage levels.

4.4 Description of Compartments and Parameters

Here, S is the number of susceptible individuals, I is the number of infected individuals, C is the number of confirmed cases, R is the number of recovered individuals. For the counterfactual analysis, V describes the number of individuals from age group 1 are not susceptible to influenza due to vaccination. Note that this is a subset of the entire vaccinated population, due to the effectiveness of the vaccine. Moreover, β is the transmission rate, f is the confirmation rate, and γ is the recovery rate.

4.5 Model Assumptions

It was assumed that individuals that recover from influenza are fully immune and do not return back to the susceptible compartment for this study. We also treat vaccine effectiveness by claiming a percentage of the vaccinated people are fully immune, and the complementary percentage are fully susceptible. The confirmed cases compartment is for individuals who have gone to the hospital and gotten tested and confirmed for the virus. Additionally, the infected compartment consists of

individuals who have not gone to the hospital while sick. The number of susceptible individuals were 791385 and 3404683 in group 1 and 2, respectively. This was calculated using the online IHDA demographics data(7).

To reduce non-identifiability issues, γ was assumed to be 1 upon discussion with experts.⁴ The recovery rate and confirmation rate f is assumed to be constant across age groups.

For the counter factual analysis, the vaccine effectiveness was assumed to be 50% and 75%, with different levels of population coverages, 20%, 30%, and 50%.

4.6 Parameter Fitting Methods

Parameters for β_1, β_2 , and f was estimated using Bayesian inference methods outlined in reference (6) for given prior values.⁵ All parameters were assumed to have a uniform prior distribution. The β matrix was estimated based upon the United Kingdom (UK) data from the POLYMOD study (8) and the population estimates data from Government of Alberta (7).⁶ In addition, compartments I01 and I02 were also estimated using this approach.

Following numerical analysis on the system of ODEs, there will be a two step simulation:

1. Estimate parameters via simulation using Monte Carlo Markov Chain (MCMC).
2. Determine accuracy of estimated Bayesian parameters using a Goodness of Fit test.

More detailed information on this numerical analysis method can be found in (6).

4.7 Software

For data cleaning, Python and R was used to filter the necessary information. MATLAB was used to conduct the Bayesian fitting and the counter factual analysis.

⁴This was a discussion I had with Dr. Betsy Varughese - previous studies have shown that the recovery rate of influenza is about 7 days.

⁵All prior distributions for the parameters were chosen to be uniform.

⁶See appendix for weighted contact matrix

5 Results

As mentioned in the methods section, parameters were simulated using MCMC and a Goodness-of-Fit test was conducted. As well, we built a case of what would happen to the number of confirmed cases if there were higher coverage of vaccinations by conducting a counterfactual analysis.

5.1 Fitted Parameters

The following table outlines the estimated parameters based on the methods mentioned previously.

Table 1: Parameter Values Fitted by Bayesian Inference Method

Parameters	β_1	β_2	f	I01	I02	p
Maximum Fitted Value	4.8257e-08	3.6462e-08	0.0032	26.3818	2.8303e+03	0.0949
Median Fitted Value	4.9057e-08	3.6495e-08	0.0032	533.8320	2.6279e+03	0.0905
95% CI	(3.8103e-08, 7.0704e-08)	(2.9668e-08, 3.7307e-08)	(0.0027, 0.0300)	(57.1337, 997.6486)	(2.0187e+03, 3.0000e+03)	(0.0120, 0.1211)

Furthermore, a Bayesian p-value was calculated to be 0.3661. This p-value was based on a chi-squared realized discrepancy function that has an approximate chi-squared distribution with n-degrees of freedom.(6)

The following plot shows how these parameters fit the data:

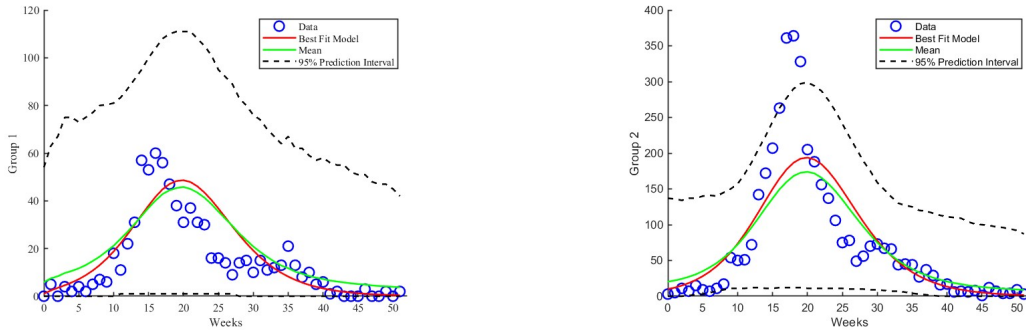


Figure 4: Plot of Fitted Data using Estimated Parameters, Left: Group 1, Right: Group 2

Qualitatively, we can see that the general pattern follows the actual data quite well for both groups. However, it is notable that the 95% prediction interval in group 2 does not capture the true data.

5.2 Counter Factual Analysis

A vaccination compartment was added to group 1 to conduct a counter factual analysis. For this, the two levels of vaccine effectiveness were assumed to be 50% and 75%. Additionally, 20%, 30%, and 50% population coverages of group 1 were assumed. The model was refitted accordingly and the cumulative counts of the counter factual analysis were compared to the cumulative counts of the exact fit during the 52-week period.

The following tables show the cumulative counts of each counter factual case for group 1 as well as the percent difference to the exact fit.

Table 2: Cumulative Counts of Counter Factual Analysis for Group 1 - 50% Effectiveness (% Decrease in red)

Model	20% Coverage	30% Coverage	50% Coverage
911	793	736	628
-	12.95%	19.21%	31.06%

Table 3: Cumulative Counts of Counter Factual Analysis for Group 1 - 75% Effectiveness (% Decrease in red)

Model	20% Coverage	30% Coverage	50% Coverage
911	736	654	500
-	19.21%	28.21%	45.11%

In addition to the plots above, confirmed cases within each compartment were also plotted for varying levels of vaccine effectiveness and coverages.

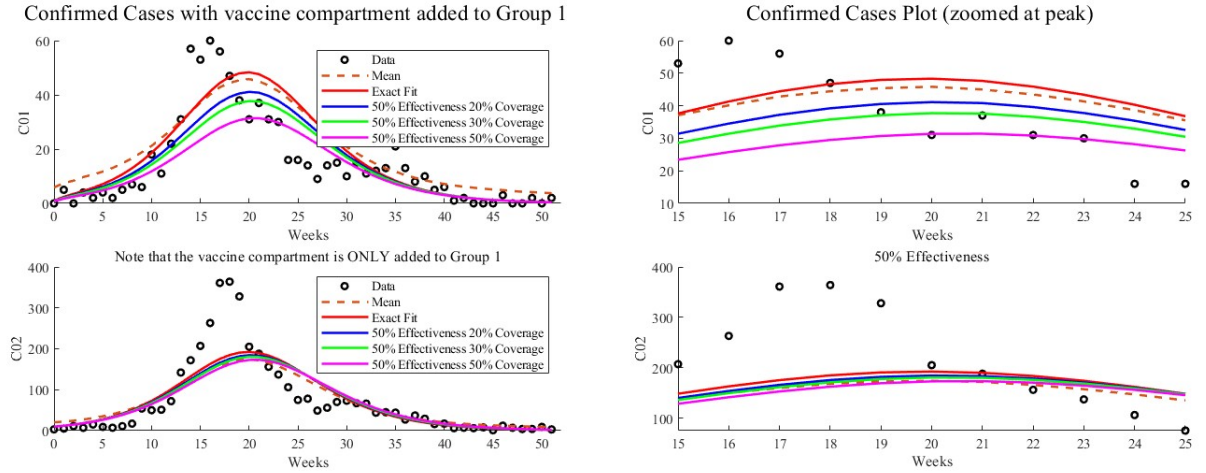


Figure 5: Plot of Cumulative Cases with 50% Vaccine Effectiveness, Left: Cumulative Cases Compartments, Right: Zoomed in at peak

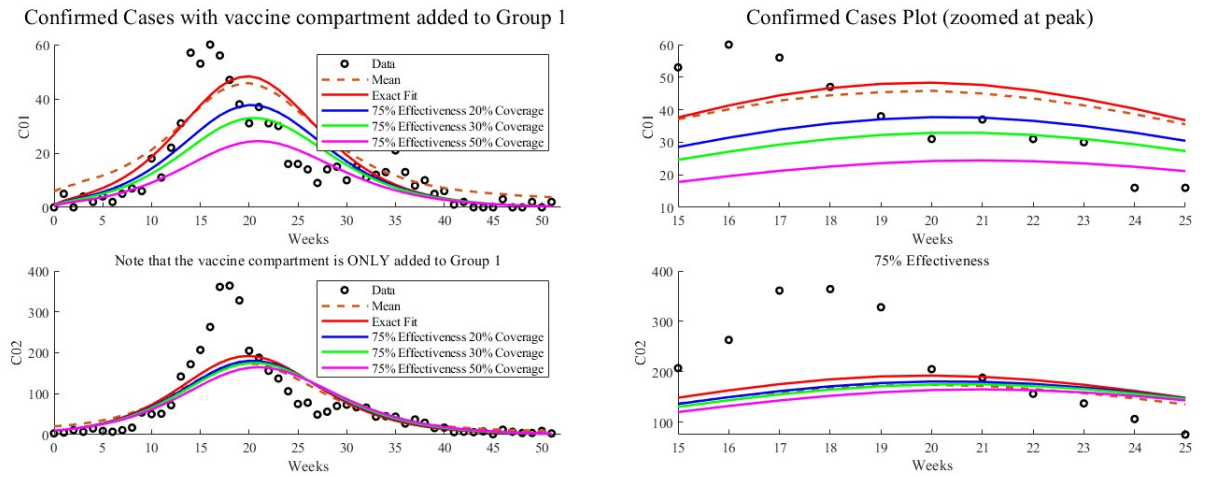


Figure 6: Plot of Cumulative Cases with 75% Vaccine Effectiveness, Left: Cumulative Cases Compartments, Right: Zoomed in at peak

As well, the following plots show cumulative cases for both age groups in addition to the plots above.

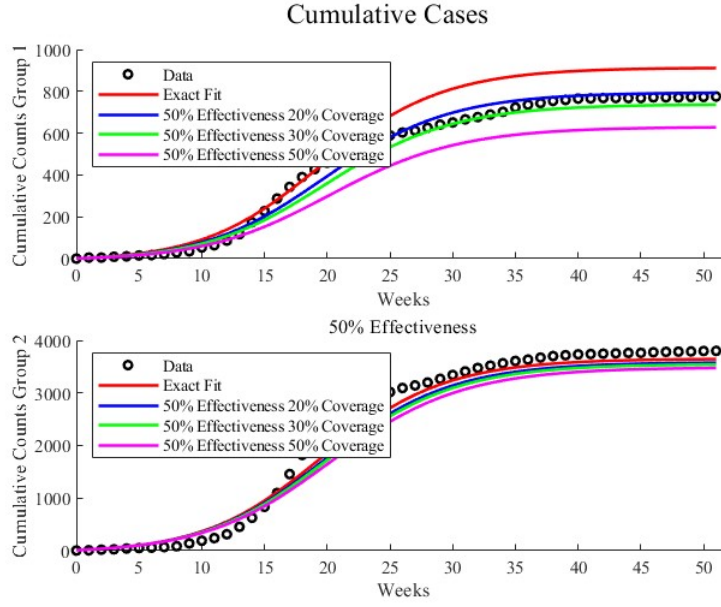


Figure 7: Plot of Cumulative Cases with 50% Vaccine Effectiveness

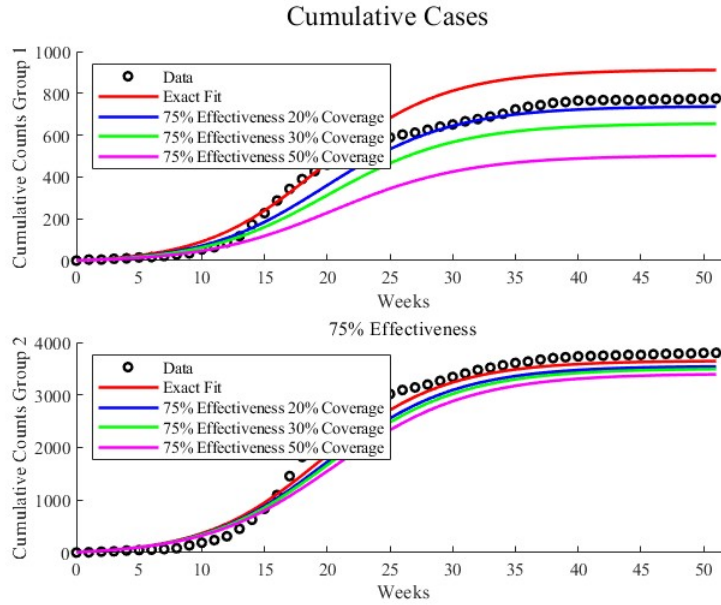


Figure 8: Plot of Cumulative Cases with 75% Vaccine Effectiveness

For both groups, it is clearly shown that increasing the percentages of vaccine effectiveness and vaccine coverage leads to a decrease in influenza cases.

6 Discussion and conclusions

For this study, count data of weekly influenza cases in 2016-2017 was used. The data was provided by Dr. Betsy Varughese. Original data consisted of many variables, however a data dictionary was not provided. Therefore, it may be beneficial to have the definitions as supplementary material as definitions of factors vary between studies.

To begin our analysis, the parameters in the influenza model were estimated using a Bayesian inference approach. Although the Bayesian p-value was evidence that the model predictions fit the data, we see in the posterior plot for the I01 compartment that there are some concerns of non-identifiability issues.⁷ Some methods to resolve this problem is to either rearrange prior values for the different parameters, find a better prior distribution, or conduct a literature search to determine a more accurate range of values for the I01 compartment.

Also, parameters γ and f were assumed to be constant across both age groups to reduce computational capacity, however it is important to note that this is not the case in the real world due to under-reporting of influenza cases. In reality, these term must be considered as a function of time. Moreover, it is not recommended for both groups to have the same recovery and confirmation rate. For example, older individuals are more likely to end up at the hospital and will get confirmed compared to younger groups, therefore assuming same case detection rate between age groups is not realistic.

A counter factual analysis was also conducted for this project, where we saw that by adding a vaccination compartment to group 1, number of cumulative cases decreased in both age groups. Group 1 showed a significant decrease in cumulative cases, highest decrease being 45.11% from the exact fit when the vaccine effectiveness was 75% and 50% of the population was assumed to receive the vaccine. Group 2 also showed a decrease in cumulative cases, which can be seen in tables 4 and 5 in the appendix. For future studies, it is recommended that a vaccination compartment should also be added to the second group to capture the dynamics of the Albertan population more accurately.

⁷See section 7.3 for plots

References

- [1] M. T. Sofonea, S. Cauchemez, and P.-Y. Boëlle, “Epidemic models: Why and how to use them,” *Anaesthesia Critical Care amp; Pain Medicine*, vol. 41, no. 2, p. 101048, 2022.
- [2] M. J. Keeling and P. Rohani, *Modeling infectious diseases in humans and animals*. Princeton University Press, 2008.
- [3] W. Smith, C. Andrewes, and P. Laidlaw, “A virus obtained from influenza patients,” *The Lancet*, vol. 222, no. 5732, p. 66–68, 1933.
- [4] “Key facts about influenza (flu),” Oct 2022.
- [5] N. Bacaër, *McKendrick and Kermack on epidemic modelling (1926–1927)*, pp. 89–96. London: Springer London, 2011.
- [6] W. C. Roda, “Bayesian inference for dynamical systems,” *Infectious Disease Modelling*, vol. 5, pp. 221–232, 2020.
- [7] “Interactive health data application (government of alberta).”
- [8] J. Mossong, N. Hens, M. Jit, P. Beutels, K. Auranen, R. Mikolajczyk, M. Massari, S. Salmaso, G. S. Tomba, J. Wallinga, and et al., “Social contacts and mixing patterns relevant to the spread of infectious diseases,” *PLoS Medicine*, vol. 5, no. 3, 2008.

7 Appendix

All code used to create plots and complete analyses can be provided upon request at sp8@ualberta.ca.

7.1 Systems of ODEs

$$S'_1(t) = -\beta_{11}S_1I_1 - \beta_{12}S_1I_2 \quad (1)$$

$$S'_2(t) = -\beta_{21}S_2I_1 - \beta_{22}S_2I_2 \quad (2)$$

$$I'_1(t) = \beta_{11}S_1I_1 + \beta_{12}S_1I_2 - fC_1 - \gamma I_1 \quad (3)$$

$$I'_2(t) = \beta_{21}S_2I_1 + \beta_{22}S_2I_2 - fC_2 - \gamma I_2 \quad (4)$$

$$C'_1(t) = fI_1 \quad (5)$$

$$C'_2(t) = fI_2 \quad (6)$$

$$R'_1(t) = \gamma I_1 \quad (7)$$

$$R'_2(t) = \gamma I_2 \quad (8)$$

7.2 Data

The following figures allow us to visualize the 2016-2017 Infleunza data.

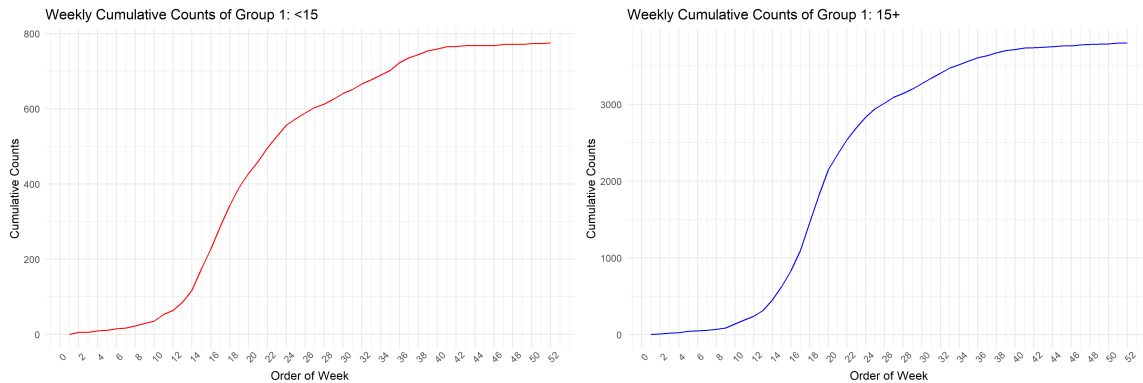


Figure 9: Cumulative Counts of Group 1 (Red) and Group 2 (Blue).

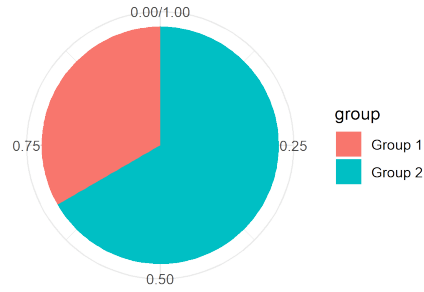


Figure 10: Proportion of Group 1 (Red) and Group 2 (Blue) in original dataset.

The weighted contact matrix was estimated using a combination of POLYMOD and IHDA data as mentioned in section 4.6:

$$\begin{bmatrix} 6.6933 & 6.4921 \\ 1.5090 & 9.4592 \end{bmatrix} \quad (9)$$

Supporting information Table S8.4. Contact matrix of all reported contacts in Great Britain consisting of the average number of contact persons recorded per day per survey participant

a) all reported contacts (physical and non-physical contacts)

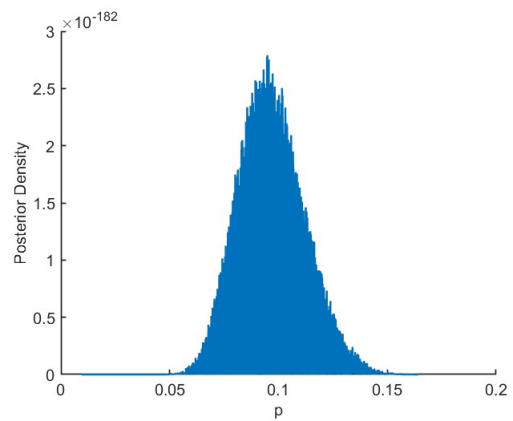
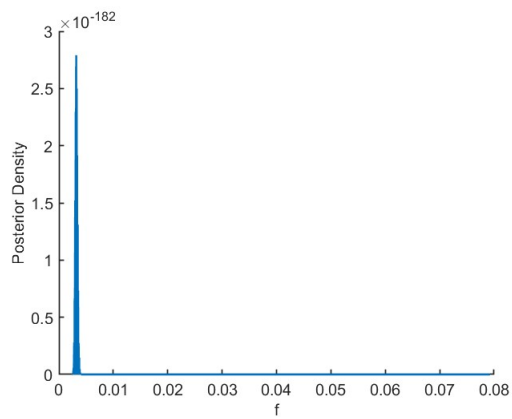
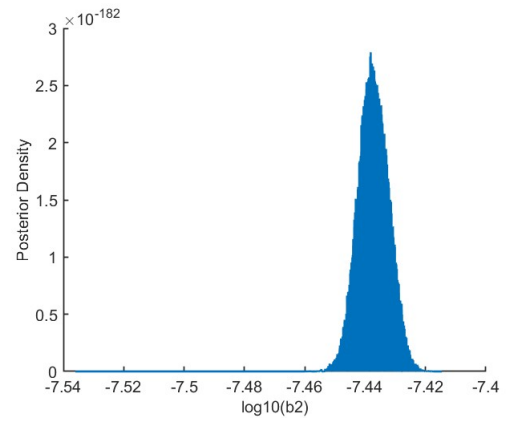
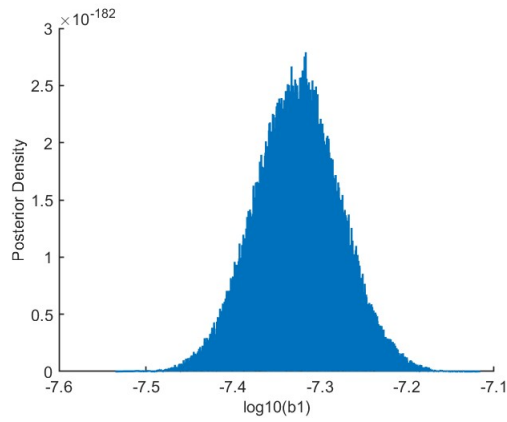
age of contact	age group of participant															
	00-04	05-09	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59	60-64	65-69	70+	
00-04	1.92	0.65	0.41	0.24	0.46	0.73	0.67	0.83	0.24	0.22	0.36	0.20	0.20	0.26	0.13	
05-09	0.95	6.64	1.09	0.73	0.61	0.75	0.95	1.39	0.90	0.16	0.30	0.22	0.50	0.48	0.20	
10-14	0.48	1.31	6.85	1.52	0.27	0.31	0.48	0.76	1.00	0.69	0.32	0.44	0.27	0.41	0.33	
15-19	0.33	0.34	1.03	6.71	1.58	0.73	0.42	0.56	0.85	1.16	0.70	0.30	0.20	0.48	0.63	
20-24	0.45	0.30	0.22	0.93	2.59	1.49	0.75	0.63	0.77	0.87	0.88	0.61	0.53	0.37	0.33	
25-29	0.79	0.66	0.44	0.74	1.29	1.83	0.97	0.71	0.74	0.85	0.88	0.87	0.67	0.74	0.33	
30-34	0.97	1.07	0.62	0.50	0.88	1.19	1.67	0.89	1.02	0.91	0.92	0.61	0.76	0.63	0.27	
35-39	1.02	0.98	1.26	1.09	0.76	0.95	1.53	1.50	1.32	1.09	0.83	0.69	1.02	0.96	0.20	
40-44	0.55	1.00	1.14	0.94	0.73	0.88	0.82	1.23	1.35	1.27	0.89	0.67	0.94	0.81	0.80	
45-49	0.29	0.54	0.57	0.77	0.97	0.93	0.57	0.80	1.32	1.87	0.61	0.80	0.61	0.59	0.57	
50-54	0.33	0.38	0.40	0.41	0.44	0.85	0.60	0.61	0.71	0.95	0.74	1.06	0.59	0.56	0.57	
55-59	0.31	0.21	0.25	0.33	0.39	0.53	0.68	0.53	0.55	0.51	0.82	1.17	0.85	0.85	0.33	
60-64	0.26	0.25	0.19	0.24	0.19	0.34	0.40	0.39	0.47	0.55	0.41	0.78	0.65	0.85	0.57	
65-69	0.09	0.11	0.12	0.20	0.19	0.22	0.13	0.30	0.23	0.13	0.21	0.28	0.36	0.70	0.60	
70+	0.14	0.15	0.21	0.10	0.24	0.17	0.15	0.41	0.50	0.71	0.53	0.76	0.47	0.74	1.47	

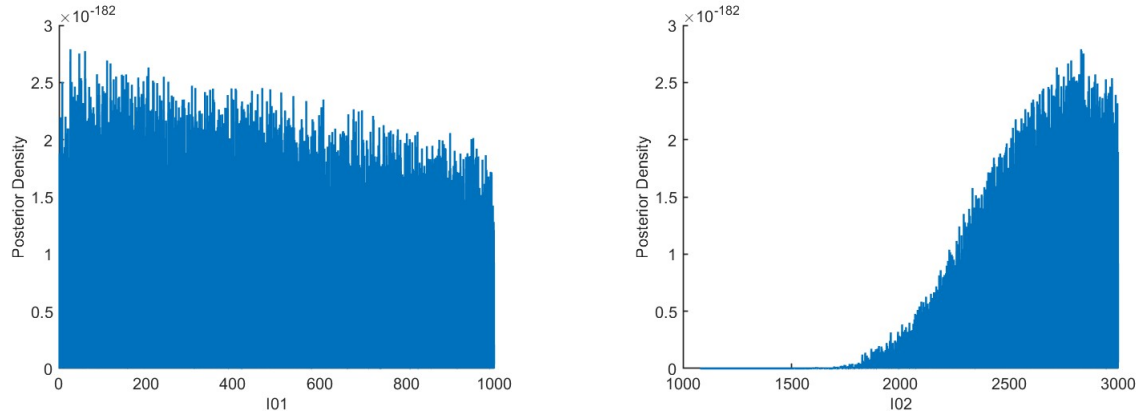
Figure 11: Contact Rates from POLYMOD study

Geography	Year	Sex	Age	Population
AB	2016	BOTH	ALL	4196068
AB	2016	BOTH	00to04	274113
AB	2016	BOTH	05to09	272631
AB	2016	BOTH	10to14	244641
AB	2016	BOTH	15to19	247030
AB	2016	BOTH	20to24	279677
AB	2016	BOTH	25to29	335868
AB	2016	BOTH	30to34	356151
AB	2016	BOTH	35to39	322538
AB	2016	BOTH	40to44	291602
AB	2016	BOTH	45to49	274584
AB	2016	BOTH	50to54	287290
AB	2016	BOTH	55to59	278838
AB	2016	BOTH	60to64	228872
AB	2016	BOTH	65to69	174507
AB	2016	BOTH	70+	327726
				3404683

Figure 12: IHDA Population Estimates in Alberta - 2016

7.3 Posterior Density Plots





7.4 Counter Factual Analysis for Group 2

Table 4: Cumulative Counts of Counter Factual Analysis for Group 2 - 50% Effectiveness (% Decrease in red)

Model	20% Coverage	30% Coverage	50% Coverage
3643	3575	3541	3474
-	1.86%	2.79%	4.64%

Table 5: Cumulative Counts of Counter Factual Analysis for Group 2 - 75% Effectiveness (% Decrease in red)

Model	20% Coverage	30% Coverage	50% Coverage
3643	3541	3491	3392
-	2.79%	4.17%	6.89%